

ZHAOMIN WU

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BIO

Dr. Zhaomin Wu is a Research Fellow at the Department of Computer Science, National University of Singapore (NUS). He earned his Ph.D. in Computer Science from NUS in 2024, advised by Prof. Bingsheng He, and obtained a Bachelor's degree from Huazhong University of Science and Technology in 2019. Dr. Wu's research focuses on trustworthy machine learning, with specific interests in trustworthy machine learning, federated learning, and multi-agent system. He received the Singapore NRF Postdoctoral Award in 2026. His work has been recognized with awards such as the SIGMOD Honorable Mention for Best Artifact (2023). His publications appear in top-tier venues including NeurIPS, ICLR, SIGMOD, KDD, WWW, ACL, EMNLP, AACL, MLSys, and TKDE.

EDUCATION

National University of Singapore (NUS), Singapore Doctor of Philosophy – Computer Science Thesis: Towards Practical Vertical Federated Learning <i>Honorable Mention for Best Ph.D. Thesis Award</i>	07/2019 – 05/2024 Advisor: Prof. Bingsheng He
Huazhong University of Science and Technology (HUST), China Bachelor of Engineering – ACM class, Computer Science and Technology GPA: 3.91/4.0, Rank: 10/281 <i>Outstanding Undergraduate Thesis Award</i>	09/2015 – 06/2019

EMPLOYMENT

National University of Singapore (NUS), Singapore	05/2024 – Present
• <i>NRF Postdoctoral Fellow</i> , Department of Computer Science Principal Investigator (PI) of an NRF-funded postdoctoral project	04/2026 – Present
• <i>Research Fellow</i> , Department of Computer Science Project PI: Prof. Bingsheng He	07/2025 – 03/2026
• <i>Research Fellow</i> , Institute of Data Science Project PI: Prof. Ng See Kiong	05/2024 – 06/2025

PUBLICATIONS

 *Equal contribution, †Corresponding author

2026

[ICLR 2026] LLM DNA: Tracing Model Evolution via Functional Representations (*Oral, 1%*)
Zhaomin Wu, Haodong Zhao, Ziyang Wang, Jizhou Guo, Qian Wang, Bingsheng He

[ICLR 2026] Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts (*Oral, 1%*)
Zhaomin Wu, Mingzhe Du, See-Kiong Ng, Bingsheng He

[ICDE 2026] WikiDBGraph: Large-Scale Database Graph of Wikidata for Collaborative Learning
Zhaomin Wu^{*}, Ziyang Wang^{*}, Bingsheng He

[KDD 2026] Mining Intrinsic Rewards from LLM Hidden States for Efficient Best-of-N Sampling

Jizhou Guo, Zhaomin Wu[†], Hanchen Yang, Philip S. Yu

[arXiv 2026] ProtegoFed: Backdoor-Free Federated Instruction Tuning with Interspersed Poisoned Data

Haodong Zhao, Jinming Hu, Zhaomin Wu[†], Zongru Wu, Wei Du, Junyi Hou, Caibei Zhao, Zhuosheng Zhang, Bingsheng He, Gongshen Liu[†]

[WWW 2026] Personalized Federated Fine-Tuning for LLMs via Data-Driven Heterogeneous Model Architectures (*Oral, 9%*)

Yicheng Zhang, Zhen Qin, Zhaomin Wu, Shuiguang Deng[†]

2025

[EMNLP 2025] Model-based Large Language Model Customization as Service

Zhaomin Wu^{*}, Jizhou Guo^{*}, Junyi Hou, Bingsheng He, Lixin Fan, Qiang Yang

[arXiv 2025] Vertical Federated Learning in Practice: The Good, the Bad, and the Ugly

Zhaomin Wu, Zhen Qin, Junyi Hou, Haodong Zhao, Qinbin Li, Bingsheng He, Lixin Fan

[ACL 2025] Federated Data-Efficient Instruction Tuning for Large Language Models

Zhen Qin, Zhaomin Wu[†], Bingsheng He, Shuiguang Deng[†]

[arXiv 2025] Disagreements in Reasoning: How a Model’s Thinking Process Dictates Persuasion in Multi-Agent Systems

Haodong Zhao, Jidong Li, Zhaomin Wu[†], Tianjie Ju, Zhuosheng Zhang, Bingsheng He, Gongshen Liu[†]

2024

[NeurIPS 2024] Federated Transformer: Multi-Party Vertical Federated Learning on Practical Fuzzily Linked Data

Zhaomin Wu, Junyi Hou, Yiqun Diao, Bingsheng He

[ICLR 2024] VertiBench: Advancing Feature Distribution Diversity in Vertical Federated Learning Benchmarks

Zhaomin Wu, Junyi Hou, Bingsheng He

2023

[SIGMOD 2023] DeltaBoost: Gradient Boosting Decision Trees with Efficient Machine Unlearning (*Honorable Mention for Best Artifact, 3/~600*)

Zhaomin Wu, Junhui Zhu, Qinbin Li, Bingsheng He

[MLSys 2023] FedTree: A Federated Learning System For Trees

Qinbin Li, Zhaomin Wu, Yanzheng Cai, Yuxuan Han, Ching Man Yung, Tianyuan Fu, Bingsheng He

2022

[NeurIPS 2022] A Coupled Design of Exploiting Record Similarity for Practical Vertical Federated Learning

Zhaomin Wu, Qinbin Li, Bingsheng He

[TBD 2022] Practical Vertical Federated Learning with Unsupervised Representation Learning

Zhaomin Wu, Qinbin Li, Bingsheng He

[TIST 2022] The OARF Benchmark Suite: Characterization and Implications for Federated Learning Systems

Sixu Hu, Yuan Li, Xu Liu, Qinbin Li, Zhaomin Wu, Bingsheng He

2021

[TKDE 2021] A Survey on Federated Learning Systems: Vision, Hype and Reality for Data Privacy and Protection

Qinbin Li, Zeyi Wen, Zhaomin Wu, Sixu Hu, Naibo Wang, Yuan Li, Xu Liu, Bingsheng He

2020

[AAAI 2020] Privacy-Preserving Gradient Boosting Decision Trees

Qinbin Li, Zhaomin Wu, Zeyi Wen, Bingsheng He

HONORS AND AWARDS

Grants

- **NRF Postdoctoral Award** 2026
Awarded by the National Research Foundation (Singapore); four-year salary support and an independent SGD 250,000 research grant.
- **Google Cloud Research Credit Award** 2026
Awarded by Google; USD 5,000 Google Cloud credits over one year.

Research Awards and Professional Recognition

- **SIGMOD Honorable Mention for Best Artifact** 2023
Awarded by ACM SIGMOD; 3 selected among ~600 submissions.
- **Honorable Mention for Best PhD Thesis Award** 2024
Awarded by the School of Computing, National University of Singapore; 5 selected among ~150 PhD graduates.
- **Best Research Staff Award** 2025
Awarded by the Institute of Data Science, National University of Singapore; 2 selected among ~50 research staff.
- **Dean's Graduate Research Excellence Award** 2023
Awarded by the School of Computing, National University of Singapore; 10 awardees among ~150 graduate researchers.
- **ICML 2026 Gold Reviewer Award** 2026
Awarded by the International Conference on Machine Learning (ICML); top 25% among reviewers.

Undergraduate Honors

- **Outstanding Students** 2016
Awarded by Huazhong University of Science and Technology; top 1%.
- **National Scholarship** 2017
Awarded by the Ministry of Education of China; top 1% at HUST.
- **Outstanding Undergraduate Thesis** 2019
Awarded by Huazhong University of Science and Technology; top 5%.

- **Merit Student** 2016
Awarded by Huazhong University of Science and Technology; top 5%.
- **Meritorious Winner, Mathematical Contest in Modeling** 2018
Awarded by COMAP; top 10% worldwide.

INVITED TALKS & KEYNOTES

2026

Invited Talk: “When Data and Models Stay Hidden: Toward Trustworthy AI Collaboration” May 2026
Université de Montréal (UdeM) and Mila; hosted by Prof. Bang Liu

Keynote: “Bridging Data Silos with Practical Federated Learning” Jan. 2026
AAAI 2026 FLCA Workshop

2025

Invited Talk: “Towards Practical Vertical Federated Learning Systems” May 2025
DASFAA 2025 Trust Day; on behalf of Prof. Bingsheng He

Invited Talk: “Bridging Private Data Silo with Machine Learning” Feb. 2025
NUS Open House 2025

PROFESSIONAL SERVICES

2026

Conferences: ICML (Reviewer), NeurIPS (Reviewer)

Journals: ACM Computing Survey (Reviewer), TPDS (Reviewer), TKDE (Reviewer), TMC (Reviewer)

2025

Conferences: ICLR (Reviewer), ICML (Reviewer), NeurIPS (Reviewer), AAAI (Reviewer), EMNLP (Reviewer)

Journals: TDSC (Reviewer), TPDS (Reviewer), TKDE (Reviewer), TAI (Reviewer)

2024

Conferences: ICLR (Reviewer), NeurIPS (Reviewer), KDD (Reviewer), AISTATS (Reviewer)

Journals: TKDD (Reviewer), JPDC (Reviewer), TNNLS (Reviewer)

2023

Conferences: NeurIPS (Reviewer), SIGMOD (Artifact Reviewer)

Journals: TPAMI (Reviewer), TKDE (Reviewer), TNNLS (Reviewer), IJCV (Reviewer), IoT-J (Reviewer)

2022

Conferences: PAKDD (Reviewer)

Journals: TPDS (Reviewer), TIST (Reviewer)

2021

Conferences: PAKDD (Reviewer)

2020

Tutorial: IJCAI (Tutorial Speaker)

TEACHING & MENTORSHIP

Teaching Assistant

Introduction to Operating Systems (CS2106)	Fall 2022
Software Testing (CS4218)	Spring 2022
Neural Networks and Deep Learning (CS5242)	Fall 2021
Programming Methodology (II) (CS2030)	Spring 2021
Big Data Systems for Data Science (CS4225)	Fall 2020
Data Structures and Algorithms (CS2040S)	Spring 2020
Programming Methodology (CS1010E)	Fall 2019

Mentored Students

Jiayi Li , master student, NUS	2025 – 2026
Ziyang Qiu , master student, NUS	2025 – 2026
Lifan Hu , undergraduate student, NUS	2025 – 2026
Qian Yu , undergraduate student, NUS	2025 – 2026
Ziyang Wang , master student, NUS	2024 – 2025
Haodong Zhao , visiting Ph.D. student, Shanghai Jiao Tong University	2024 – 2025
Zhen Qin , visiting Ph.D. student, Zhejiang University <i>Current: Assistant Professor, Zhejiang University</i>	2024 – 2025
Jizhou Guo , undergraduate student, Shanghai Jiao Tong University	2024
Junyi Hou , master student, NUS <i>Current: Ph.D. student, NUS</i>	2023 – 2024
Junhui Zhu , master student, NUS	2022

LANGUAGE

Chinese (Mandarin)	Native speaker
English	Professional working proficiency